

FDDT Case Report

Fraud Document Detection Tool — system-assisted analysis for human review. See Section 6 for what this report is and is not.

1. Case Details

Case ID	CASE-01673CC6 01673cc6-64f3-4f84-9eaf-c34aba7e14c0
Submitted by	test1Reviewer L2
Submission date	2026-10-07 11:35 UTC
Documents	1 Types: Other
Case type	Vendor invoice
Status / decision	Pending manual review / Pending review
Report generated	2026-10-07 11:52 UTC by testreviewer L! (Reviewer L1)
Report ID	10343ac8-e373-4fc3-b92f-79cf734e7dad

2. Executive Summary

HIGH RISK **Score 85 / 100** (computed 2026-10-07 11:49 UTC)

Why: Driven mainly by (40 points): 'ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134a.pdf': 1 page(s) are near-identical to previously submitted documents. Page 1 matches page 1 of "...

Decision status: Pending review.

Exceptions: 5 item(s) did not pass, 5 of them shown as highlighted regions on the source pages. Section 5 lists each one; Section 9 shows them on the documents.

This summary is deliberately short. The reasons behind the tier are in Section 3, the complete checklist is in Section 4.

3. Explainable Findings

The risk engine is a transparent, weighted rule set. Each row is one rule that fired, with the reason text stored at scoring time. Points are summed (raw 85), capped at 100, and mapped to a tier (medium from 30, high from 60) — here **85/100**, high.

Severity	Points	Check	Document / page	What
HIGH	40	Duplicate detection	ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134a.pdf · page 1	Sub CAS 'ff570 aeb1- 1 pag "case the tv Highl
HIGH	30	Error level analysis (ELA)	ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134a.pdf · page 1	Smc 'ff570 aeb1- page Highl
MEDIUM	15	Field validation	ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134a.pdf · page 1	Inva AE9 1312 'ff570 aeb1- Invali numb Highl

4. All Checks

The completeness record: every check that ran on every document, including those that passed. Section 3 is the narrative; this is the checklist.

Document 1 — ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134a.pdf (Other · 4 passed, 4 flagged)

Check	Result	Details
Metadata forensics PDF metadata, edit history and incremental-save markers.	Pass	No issues found
Font consistency Text set in a different font family from the text around it (text layer, or OCR font recognition on scans).	Review	Shown for review, not scored: Different font (scan, estimated) - Different font (scan, estimated) — at look like humanist sans; the text around them is grotesque sans 1 of 1 page(s) analysed (1 scanned, using OCR font recognition); 2 font families: grotesque sans, humanist sans.
Error level analysis (ELA) Pixel-level recompression differences that can indicate edited regions.	Flag	1 issue: Smoothing detected Smoothing detected — possible smoothing to hide edit traces, p.1
Copy-move detection Regions of a page that were duplicated elsewhere on the same page.	Pass	No issues found
Deleted / replaced content (ghost text) On a scan converted to editable text: faint traces of the original text left in the scanned background, with no live text on top (deleted) or running on past it (shortened).	N/A	Not applicable — no page is a scan converted to editable text
Visual inconsistency / AI-generation review Model review of fonts, alignment, contrast, sharpness, lighting and signs of AI generation.	Pass	No issues found A vision model's judgment, run twice per page — a probabilistic signal, not pixel analysis.
Signature / stamp detection Whether a signature or stamp mark is present where this kind of document normally carries one.	Pass	Found 1 signature and 1 stamp, p.1 - Stamp names the issuer. - Stamp reads: 'ENEC 12 P.O. Box 112010 Abu Dhabi United Arab Emirates Emirates Nuclear Energy Corporation'
Duplicate detection Perceptual-hash comparison of each page against previously submitted pages.	Flag	1 issue: Same file uploaded before Same file uploaded before — identical to 'case11School educational document.pdf' in CASE-2731D0C8
Field validation Date, subtotal / tax / total and reference-number plausibility.	Flag	1 issue: Invalid IBAN / TRN - Invalid IBAN / TRN — Invalid payment identifier(s): IBAN AE900030013124242920001: does not contain the printed account number 1312422920001. - Passed: date, no re-scan, reference number, date vs file. - Nothing to compare for: date order, TRN on tax invoice, invoices in the file, each invoice, scanner line, line items, tax rate, total and tax, installments, amount in words, months charged, subtotal.
Issuer verification Issuer name matched against the issuer registry.	Not checked	Not checked — the issuer registry has no active entries for in AE, so there was nothing to compare this issuer against.

5. Exceptions

Everything that did **not** pass, drawn from Section 4 and written as concrete items (5 in total). A region ID (R1, R2, ...) points to the highlight that shows the exception on the source page in Section 9 — click it to jump there.

No.	Exception	Where
1	Different font (scan, estimated) — at look like humanist sans; the text around them is grotesque sans Page 1 (scanned): 'at' looks like a different font (humanist sans) from the text around it (grotesque sans), per OCR font recognition — it may have been typed in or pasted before the page was printed or scanned. Only 1 word(s) on this page look like this font, so it is shown for review but not scored: OCR font recognition alone is not reliable on a word or two.	Document 1 (ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376idd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134a.pdf) · page 1
2	Smoothing detected — possible smoothing to hide edit traces, p.1 Page 1: possible anti-forensic smoothing detected (noise variance=4.47) — could indicate an attempt to hide edit traces.	Document 1 (ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376idd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134a.pdf) · page 1
3	Same file uploaded before — identical to 'case11School educational document.pdf' in CASE-2731D0C8 Page 1 matches page 1 of "case11School educational document.pdf" in case CASE-2731D0C8, submitted by testreviewer LI, and the two files are byte-for-byte identical (same SHA-256) — the exact same file re-uploaded.	Document 1 (ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376idd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134a.pdf) · page 1
4	Invalid IBAN / TRN — Invalid payment identifier(s): IBAN AE900030013124242920001: does not contain the printed account number 1312422920001. Invalid payment identifier(s): IBAN AE900030013124242920001: does not contain the printed account number 1312422920001.	Document 1 (ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376idd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134a.pdf) · page 1
5	Missing required field: Total amount was not found on this document	Document 1 (ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376idd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134a.pdf)

1 exception(s) show “—” because they have no place on a page (for example a missing field, or a check that did not localize its finding).

6. Limitations and Assumptions

Please read this section before relying on anything in this report.

Limitations of This Analysis

What this report is

This is **system-assisted analysis to support human review**. It is not a certified forensic or legal determination that any document is genuine, forged or fraudulent, and it must not be presented as one.

Three kinds of finding

Deterministic pixel and structure checks — error level analysis, copy-move detection and PDF metadata forensics — are computed by fixed algorithms over the file’s pixels and structure. Localized findings are drawn with **solid** boxes (ELA in red, copy-move in orange). They are signals, not proof: ELA in particular can false-positive on legitimately recompressed scans and can miss print-and-rescan tampering.

Deterministic rule-based field checks — field validation and cross-document consistency — compare extracted values by fixed rules and are drawn with **solid purple** boxes. They are exact about the values they compare, but only as reliable as the OCR and extraction that produced those values.

Model-judgment checks — the visual inconsistency review, the AI-generation assessment and the signature/stamp comparison — are **probabilistic assessments** made by a vision-language model, not deterministic proof. They can vary from run to run, and their boxes are only approximate; they are the **dashed** boxes captioned “approximate” throughout this report. Signature and stamp detection shows presence and placement only and is not an identity match; the AI-generation assessment is experimental, and its accuracy on scanned business documents has not been validated.

What the system can and cannot see

The system evaluates the **submitted document file itself**. It cannot verify facts external to the document — for example whether a real-world transaction actually occurred, whether the named issuer really produced the document, or (without a pre-existing reference signature) who a signatory is. Printed Arabic and English text are read; handwritten Arabic is not reliably read.

Absence of a finding

The absence of a flagged finding does not guarantee that a document is genuine. It means only that no supported check detected an anomaly.

Check coverage on this case

Not every check applies to every document or case. This is what actually ran here:

Check	Status	Coverage on this case
Metadata forensics	Ran	Ran on 1 of 1 document(s); 0 flagged.
Font consistency	Ran	Ran on 1 of 1 document(s); 0 flagged.
Error level analysis (ELA)	Ran	Ran on 1 of 1 document(s); 1 flagged.
Copy-move detection	Ran	Ran on 1 of 1 document(s); 0 flagged.
Deleted / replaced content (ghost text)	Not applicable	Not applicable: no page is a scan converted to editable text.
Visual inconsistency / AI-generation review	Ran	Ran on 1 of 1 document(s); 0 flagged.
Signature / stamp detection	Ran	Ran on 1 of 1 document(s); 0 flagged.
Duplicate detection	Ran	Ran on 1 of 1 document(s); 1 flagged.
Field validation	Ran	Ran on 1 of 1 document(s); 1 flagged.
Issuer verification	Ran	Ran on 1 of 1 document(s); 0 flagged.
Signature comparison (advisory)	Did not run	Did not run — no reference signature was set for this case.
Cross-document consistency	Not applicable	Not applicable — this case has a single document.

Risk score and rule versions

The risk score in this report (**85/100, high**) was computed at 2026-10-07 11:49 UTC (assessment 21a44552-ce61-4bbd-

a753-06c8a747b1e1) from the risk-rule versions that fired: `duplicate.cross_case_match v1` (weight 40); `ela.anti_forensic_signal v1` (weight 30); `field.iban_trn_validation v1` (weight 15). Tier thresholds at that time: medium from 30, high from 60.

Rule weights are versioned and are never re-applied retroactively: if an administrator later edits a rule, this case is not re-scored, and **this report remains an accurate historical record** of what was known and weighted when it was generated. A report generated later may legitimately differ.

Assumptions

What the analysis took to be true without independently verifying it. These are specific to what ran on this case.

- The issuer registry (0 active entries) was assumed to be current and accurate as of 2026-10-07 11:52 UTC. An issuer that is absent from it is reported as “not in registry”, which is a prompt to verify the issuer — not a finding that the issuer is fraudulent.
- Field values were read by OCR followed by language-model extraction and are assumed to have been read correctly. Each field carries a confidence and an “uncertain” flag. This build records that flag but does not automatically route low-confidence documents to manual review, so it is listed here for the reviewer instead: no field was marked uncertain on this case.
- Purple field highlights were located by matching extracted values back onto OCR word positions; treat their position as approximate. The comparison behind each one uses the stored, normalized values, not the highlight.
- The SHA-256 chain of custody shows the stored file is unchanged since upload. It does not show that the uploaded file is the issuer’s original — a document altered before upload is judged on its content, like any other.
- Risk-rule weights, severities and tier thresholds are administrator-configured and assumed to be appropriately calibrated; the score is advisory and is not a probability of fraud.

7. Audit Trail

Chronological case-level events, read from the append-only audit log (times in UTC). This report does not write to or duplicate the log.

Time	Event	Actor	Detail
2026-10-07 11:35 UTC	Case created	test1Reviewer L2 (Reviewer L2)	
2026-10-07 11:35 UTC	Document uploaded	test1Reviewer L2 (Reviewer L2)	ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134-SHA-256 4753e0376fdd7679...
2026-10-07 11:35 UTC	ELA, copy-move and ghost-content checks completed	System	ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134
2026-10-07 11:35 UTC	Duplicate detection completed	System	ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134
2026-10-07 11:36 UTC	Signature / stamp detection completed	System	ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134
2026-10-07 11:36 UTC	Visual inconsistency review completed	System	ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134
2026-10-07 11:49 UTC	Metadata forensics completed	System	ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134
2026-10-07 11:49 UTC	Font consistency check completed	System	ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134
2026-10-07 11:49 UTC	Text extraction, classification and field extraction completed	System	ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134
2026-10-07 11:49 UTC	Field validation and issuer verification completed	System	ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134
2026-10-07 11:49 UTC	Risk score computed	System	tier high, score 85
2026-10-07 11:49 UTC	Case status changed	System	submitted → pending_manual_review; automated checks complete

8. Appendix

The detailed technical record, for someone auditing the analysis itself rather than a first-pass reviewer.

8.1 Extracted fields

Normalized values as stored at extraction time. “Located” is the page the value was matched to (used for the purple highlights); a blank means no position was recorded. Arabic values are shown right-to-left, numbers left-to-right.

Document 1 — ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134a.pdf

Field	Value	Confidence	Located
Issuer	Emirates Nuclear Energy Corporation مؤسسة الإمارات للطاقة النووية	95%	p.1
Reference number	SC20231027/23051672/00001631	95%	p.1
Date	2023-12-20	95%	p.1
Amount	Not found	uncertain	
Subtotal	Not found	uncertain	
Tax rate	Not found	uncertain	
Tax amount	Not found	uncertain	
Document title	Salary Certificate	95%	p.1
Employee name	Nagasurya Nageshwaran Nageswaran	95%	p.1
Nationality	Indian	95%	p.1
Date of join	10 Jan 2023	95%	p.1
Employee id	00001631	95%	p.1
Work location	BNPP - Barakah	95%	p.1
Total fixed salary	44094.75 AED	95%	p.1
Total variable salary	6300.00 AED	95%	p.1
Total lump sum	50394.75 AED/Month	95%	p.1
Position	IT Engineer	95%	p.1
Grade	18	95%	p.1
Shift pattern	MON2FRI	95%	p.1
Bank name	Abu Dhabi Commercial Bank	95%	p.1
Bank account number	1312422920001	95%	p.1
Iban	AE900030013124242920001	95%	p.1
Issuer signature name	Ahmed Jasem AlShamsi Enterprise Chief Human Capital Officer	90%	p.1

8.2 Document hashes (chain of custody)

SHA-256 of each original as recorded when it was uploaded to immutable storage, re-checked against the stored file when this report was generated.

Document	SHA-256
ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134a.pdf 122,398 bytes	4753e0376fdd7679'

8.3 Technical parameters used by each check

Page rendering (input to ELA, copy-move, duplicate, visual review)

Render resolution	200 DPI
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Colour model	RGB, converted to BGR for OpenCV
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Metadata forensics

Libraries	pikepdf / PyMuPDF (PDF structure, XMP, incremental updates)
Modified-after-created rule	flagged if ModDate is more than 1 second(s) after CreationDate
Editing-software names matched	photoshop, gimp, illustrator, affinity, coreldraw, paint.net, pixlr, canva, inkscape

Error level analysis (ELA)

JPEG recompression quality	75
Error gain (scale)	50 (sqrt error term; the reference tool's default)
Contrast stretch / LUT	not applied — the raw error magnitude is thresholded directly
Region threshold	page-relative percentile 99.7 with an absolute floor of 60/255
Region area accepted	0.2% – 60% of the page
Applies to	pages with an embedded raster image only (born-digital text pages are exempt)

Copy-move detection

Keypoint algorithm	BRISK (OpenCV), detector threshold 200
Response / matching / distance thresholds	90 / 20 / 15
Minimum cluster size	30 matching keypoint pairs
Minimum region area	2% of the page
Keypoint / match caps	4,000 / 4,000

Duplicate detection

Hash	perceptual hash (imagehash pHash), 64-bit, one per rendered page
Hamming-distance threshold	5 (≤ this is flagged)
Compared against	every page hash of every previously submitted document

Visual inconsistency / AI-generation review

Model	gpt-4.1-mini
Independent runs per page	2
A finding is kept if	both runs report it, or one reports it with high confidence
Image sent to model	page render, long edge ≤ 1536px

Signature / stamp detection and comparison

Model	gpt-4.1-mini
Pages analysed per document	first 10
Output	presence and placement only; comparison is a qualitative verdict, never a numeric score

Field validation

Sub-checks	date not in the future; subtotal vs line items; total vs subtotal + tax; reference-number format
Amount tolerance	1% of the amount, minimum 0.01

Issuer verification

Matching	rapidfuzz weighted ratio against the active issuer registry, with a model-judged fallback for cross-script names
Similarity threshold	85

Field locations (purple highlights)

Method	extracted values matched back onto OCR word positions (Azure Document Intelligence Layout)
Repeated values	resolved by nearby keywords (“total”, “subtotal”, “VAT”, “date”...)
Accuracy	approximate; a field that cannot be matched is reported without a highlight



9. PDF Highlighted Regions

Every flagged page, rendered from the original with its findings drawn in. These are copies made for this report; the original files are untouched. Only pages with at least one highlight appear. Each highlight has a region ID that Section 5 refers back to.

Colour key

	ELA (solid red)
	Copy-move (solid orange)
	Font mismatch in the text layer (solid fuchsia)
	Ghost text: deleted or shortened content (solid teal)
	Field exception (solid purple)
	Model judgment (dashed blue, approximate)
	Page-level finding (not localized)

Region index

Region	Document	Page	What it shows
R1	Document 1 (ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134a.pdf)	1	Font mismatch — text set in a different font family (scan, estimated)
R2	Document 1 (ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134a.pdf)	1	ELA — possible anti-forensic signal
R3	Document 1 (ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134a.pdf)	1	Visual review — resolution / sharpness consistency (approximate)
R4	Document 1 (ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134a.pdf)	1	Duplicate detection — near-duplicate page
R5	Document 1 (ff570d85-61d4-4d32-aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134a.pdf)	1	Field exception: IBAN (does not contain the printed account number 1312422920001)

ff570d85-61d4-4d32-

aeb1-4e8e84919c3c_4753e0376fdd767972a5c1c0080680cd38ad8969a1cbe679b07020f34565134a.pdf — page 1

Section 9 · Document 1 of 1 · regions R1, R2, R3, R4, R5

Font mismatch in the text layer (solid fuchsia)

Field exception (solid purple)

Model judgment (dashed blue, approximate)

Page-level finding

مؤسسة الإمارات للطاقة النووية
Emirates Nuclear Energy Corporation

Reference Number : SC20231027/23051672/00001631

20 Dec 2023

Emirates NBD
Abu Dhabi - United Arab Emirates

R1 · Font mismatch — text set in a different font family (scan, estimated)

This is to certify that under mentioned person is currently employed at Emirates Nuclear Energy Corporation located in Emirate of Abu Dhabi.

Name	: Nagasurya Nageshwaran Nageswaran
Nationality	: Indian
Date of Join	: 10 Jan 2023
Employee ID	: 00001631
Work Location	: BNPP - Barakah
Total Fixed Salary	: 44094.75 AED
Total Variable Salary	: 6300.00 AED
Total Lump Sum	: 50394.75 AED/Month
Position	: IT Engineer
Grade	: 18
Shift Pattern	: MON2FRI

R5 · Field exception: IBAN (does not contain the printed account number 1312422920001)

1312422920001, AE900030013124242920001

This certificate has been issued upon the person's request without any liability or obligation whatsoever on the part of Emirates Nuclear Energy Corporation. The certificate shall remain valid for thirty calendar days from the date of issue. Any alteration or modification to this certificate shall render the certificate null and void.

Yours Sincerely

R3 · Visual review — resolution / sharpness consistency (approximate)

Ahmed Jasem AISHamsi
Enterprise Chief Human Capital Officer

Variable allowances are subject to change

Emirates Nuclear Energy Corporation
12

R1. Font consistency

Page 1 (scanned): 'at' looks like a different font (humanist sans) from the text around it (grotesque sans), per OCR font recognition — it may have been typed in or pasted before the page was printed or scanned. Only 1 word(s) on this page look like this font, so it is shown for review but not scored: OCR font recognition alone is not reliable on a word or two.

R2. Error level analysis (ELA) (page-level — not localized to a region)

Page 1: possible anti-forensic smoothing detected (noise variance=4.47) — could indicate an attempt to hide edit traces.

R3. Visual inconsistency / AI-generation review

Page 1 — Resolution / sharpness consistency: The signature and the circular stamp at the bottom right are noticeably softer and less sharp compared to the printed text, consistent with pen and stamp ink. This is a vision-model judgment, not pixel-level analysis — a probabilistic signal, not a definitive finding. Not counted: it is about the detected signature (covers 100% of it) — pen strokes and rubber-stamp ink ar...

R4. Duplicate detection (page-level — not localized to a region)

Page 1 matches page 1 of "case11School educational document.pdf" in case CASE-2731D0C8, submitted by testreviewer LI, and the two files are byte-for-byte identical (same SHA-256) — the exact same file re-uploaded.

R5. Field validation

Invalid payment identifier(s): IBAN AE900030013124242920001: does not contain the printed account number 1312422920001. (IBAN: AE900030013124242920001)